

Artificial intelligence and machine learning

Project Documentation format

1. Introduction

- **Project Title: OptiCrop:** Smart Agricultural Production Optimization Engine

- **Team Members:**

Team Leader : Manjusha Pachala

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Team member : Sake Sindhu

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2. Project Overview

- **Purpose:** OptiCrop is a smart agricultural production optimization system that helps farmers choose the most suitable crop based on soil and environmental conditions. The project uses machine learning to analyze parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to improve crop productivity and support better farming decisions.

Features:

Crop recommendation using Machine Learning.

- Soil nutrient analysis (N, P, K).
- Weather parameter analysis.
- User-friendly interface.
- Fast and accurate crop prediction.
- Secure storage of user data in MongoDB.
- Responsive web application.

3. Architecture

- **Frontend:** The frontend is developed using React.js. It provides an interactive interface where users can enter agricultural parameters and receive crop recommendations. React components handle user inputs, API requests, and display prediction results.

- **Backend:** The backend is built using Node.js and Express.js. It processes requests from the frontend, communicates with the machine learning model, validates user inputs, and returns crop recommendations through REST APIs.

Database: MongoDB is used to store user information, prediction history, and application data. Mongoose is used as the Object Data **Modeling**

(ODM) library for database operations.

4. Setup Instructions

• **Prerequisites:** ☐ Node.js (v18 or later)

☐ MongoDB

☐ npm

☐ Git

☐ VS Code

Installation:

Clone Repository: git clone <https://github.com/username/OptiCrop.git>

Navigate to Project: cd OptiCrop

Install Frontend Dependencies:

cd client

npm install

Install Backend Dependencies:

cd ../server

npm install

Configure Environment Variables:

PORT=5000

MONGO_URI=your_mongodb_connection_string

JWT_SECRET=your_secret_key

5. Folder Structure

OptiCrop/

|

└─ client/

```
|   ├── public/
|   ├── src/
|   |   ├── components/
|   |   ├── pages/
|   |   ├── services/
|   |   ├── App.js
|   |   └── index.js
|
|── server/
|   ├── controllers/
|   ├── models/
|   ├── routes/
|   ├── middleware/
|   ├── config/
|   ├── server.js
|   └── package.json
|
|── README.md
└── package.json
```

6. Running the Application

Start Backend

```
cd server
```

```
npm start
```

Backend runs at:

`http://localhost:5000`

Start Frontend

`cd client`

`npm start`

Frontend runs at:

`http://localhost:3000`

7. API Documentation

User Registration

POST

`/api/auth/register`

Request

```
{  
  "name": "John",  
  "email": "john@example.com",  
  "password": "123456"  
}
```

Response

```
{  
  "message": "User registered successfully"  
}
```

User Login

POST

/api/auth/login

Response

```
{  
  "token": "JWT_TOKEN"  
}
```

Crop Prediction

POST

/api/predict

Request

```
{  
  "N": 90,  
  "P": 42,  
  "K": 43,  
  "temperature": 22,  
  "humidity": 80,  
  "ph": 6.5,  
  "rainfall": 210  
}
```

Response

```
{
```

```
"recommendedCrop":"Rice"
```

```
}
```

Prediction History

GET

/api/history

8. Authentication

Authentication is implemented using JSON Web Tokens (JWT). Users register and log in securely. After successful login, a JWT token is generated and sent to the client.

Protected API routes require a valid JWT token for authorization. Passwords are securely encrypted before storage in MongoDB.

9. User Interface

The application includes:

- Login Page
- Registration Page
- Home Dashboard
- Crop Prediction Form
- Prediction Results Page
- User Profile
- Prediction History

10. Testing

The application includes:

- Login Page
- Registration Page
- Home Dashboard
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- Prediction Results Page
- User Profile
- Prediction History

11. Screenshots or Demo

12. Known Issues

- ☐ Predictions depend on the quality of the input dataset.
- ☐ Internet connection is required for online deployment.
- ☐ Large numbers of simultaneous requests may slightly

increase response time.

13. Future Enhancements

- ☐ AI-powered fertilizer recommendation.
- ☐ Disease detection using leaf image analysis.
- ☐ Weather forecasting integration.

- ☐ Real-time IoT sensor integration.
- ☐ Multi-language support.
- ☐ Mobile application for Android and iOS.
- ☐ Voice-based farming assistant.
- ☐ Market price prediction for crops.